Which asset adds the most diversification?

Ballast engine addendum - 8 Nov 2026 - private research, T-bill registry, $100M, $250k/month x120 (labeled beyond the confirmed year), 10-year horizon, property fixed 10%. Each row: the candidate added to the current universe, sleeve share from the frozen 1-year optimizer ("tilt" = exploratory 10% sleeve allocation where the optimizer declined it), then the validated 10-year mix at the 6% cap (current) and the 40% cap (owner default, 8 Nov). CE = certainty equivalent per year - the judge's safe-spending number. Deltas are vs the control on the SAME fit window.

Asset added	Fit window	Sleeve share (1y optimizer)	6% cap: T-bills, CE (delta)	40% cap: T-bills, CE (delta)
Nothing (current universe)	2005-2026	-	80%, $96.8M	42%, $109.1M
Managed futures (DBMF)	2019-2026 *	41%	68%, $121.3M (+9.2)	10%, $142.1M (+20.4)
TIPS (TIP)	2005-2026	6%	72%, $99.8M (+3.0)	38%, $110.8M (+1.8)
Broad commodities (DBC)	2006-2026	10% (tilt)	72%, $98.3M (+3.9)	42%, $107.3M (+1.0)
US aggregate bonds (AGG)	2005-2026	10% (tilt)	72%, $99.3M (+2.5)	32%, $111.2M (+2.1)
20+yr Treasuries (TLT)	2005-2026	14%	80%, $96.8M (-0.0)	35%, $111.5M (+2.5)
US REITs (VNQ)	2005-2026	10% (tilt)	78%, $97.3M (+0.5)	50%, $107.1M (-2.0)

* Managed futures only has history from May 2019, so its whole universe is fitted on 2019-2026 - no 2008, no 2011, no 2013. That window is calm for stocks and strong for trend-following (2022), which flatters every number on its row; its deltas are honest (vs the same-window control) but the window is not.

Reading the answer

1. Managed futures (DBMF/KMLM type) is the strongest addition the data supports. Near-zero or negative correlation to the current mix, including in the worst equity weeks (-0.33 for KMLM overall), the largest diversification-metric gains (effective bets 4.2 -> 5.1), and the biggest CE gains at both caps. Caveat above is the reason to size it carefully, not to skip it: 7.5 years of ETF history is thin. A longer index series (e.g. SG Trend) is the follow-up.

2. TIPS (TIP) is the robust runner-up. Full 2005-2026 history, picked by the optimizer on its own (6% of the sleeve), helps at both caps, and adds the one exposure the universe lacks: inflation-linked government bonds. Screening: correlation 0.52 to the mix, effective bets 4.2 -> 4.6.

3. Commodities and aggregate bonds are marginal. Both need a forced tilt; commodities help only at the strict cap (2022-driven), aggregate bonds overlap with the IEF sleeve (corr 0.74). Long Treasuries only pay at the loose cap; REITs subtract at the loose cap. Rejected at screening: developed ex-US equity (VEA) and low-vol (USMV) - too correlated to add anything; SCHP/PDBC/KMLM duplicates with shorter history.

The portfolio under the new defaults (10y, 40% cap)

Validated mix: **42.5% T-bills, 22.5% US equity, 14.4% 7-10yr Treasuries, 10.7% gold, 10% property** - CE $109.1M/yr, median leftover $140M, 5% bad case $93M, expected return 5.9%/yr, every payment funded on every final path, independent check passes. Finding: at the 40% cap the 15% year-10 loss limit now BINDS - relaxed to 40% it would allow 12.5% bills (CE higher, 5% bad case lower). The binding constraint has flipped from the any-point dip to the end-of-decade outcome.

Method and honesty notes: mix search over T-bill share (2.5% steps, risky sleeve frozen at 1y-optimizer proportions), 4x200 fresh validation paths, 4x400 independent final paths, one-step-safer walk-back; nominal dollars; fitted 1.8% average bill yield (today's ~4% would lift every row); not a full 10-year cvxpy optimization; screening measures: correlation in worst-5% equity weeks, Meucci effective number of bets, Choueifaty-Coignard diversification ratio. Code/results: branch owner-defaults-10y-cml40, studies/tbill/study_diversification.py.